

# Areg Hovumyan

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## Education

California State Polytechnic University, Pomona  
Bachelor of Science in Computer Engineering — GPA: 3.6/4.0

Pomona, CA  
Expected Graduation December 2027

## EXPERIENCE

### C&DH Subsystem Lead: L'SPACE (Academy)

Remote

NASA's student acceleration program - Robotic Lunar Rover Mission

May 2026 – September 2026

- Designed the lunar rover's end-to-end Communication and Data Handling (C&DH) architecture by communicating with mechanical and electrical teams; presented it to NASA engineers and improved the design based on their feedback from the PDR/CDR.
- Defined the Earth-Moon communication concept and autonomous fault recovery through NeoPan, adopted NASA's F Prime (F) flight software framework to meet mission requirements.
- Supported design reviews, hardware-in-the-loop tests, and bench testing; authored subsystem documentation used to onboard incoming members.

### Software Engineer: Autonomous Vehicle Lab (Research)

Pomona, CA

University Research Lab

April 2026 – September 2026

- Built an autonomous interceptor drone with onboard object detection integrated into the flight controller, delivering a flight-ready prototype under a \$2,000 budget and 93% accuracy.
- To enable drive-by-wire steering, built a closed-loop stepper controller (CL57T) on a microcontroller with CAN telemetry over a TJA1051 transceiver, accepting PC input 75 Hz / 100 ms latency.
- Traced inconsistent I2C failures on custom PCBs to missing pull-ups and SPI clock-polarity errors using an oscilloscope and logic analyzer; using AL, wrote bare-metal C/C++ drivers to validate fixes, cutting board bring-up from 2 weeks to 3 days.
- Engineered a GitHub Actions CI/CD pipeline running unit tests on every pull request, catching 4 regressions before merge.
- Automated startup, deployment, and process management with systemd services and Bash scripts, reducing system setup from 6 min to 3 min.

## PROJECTS

### Software Engineering Team Lead: Autonomous Hexacopter (University Club)

Pomona, CA

Lockheed Martin sponsored project

June 2025 – Present

- Developing a ROS 2 bridge to Anduril Lattice that fuses MAVLink telemetry with onboard perception, geolocates detections in WGS-84 within 5 m, and publishes them as Lattice track entities at 10 Hz.
- Built ROS 2 mission nodes for object detection (0.60 mAP@0.5), area mapping, 4K image processing (under 0.7 s/image), and autonomous payload release over target.
- Gave 8 student-engineers a consistent dev environment, containerized the ROS 2 stack with Docker and introduced CI/CD, test-first development, and pair programming, reducing integration defects by 13%.
- Rebalanced CPU/GPU load between detection and image capture with the Electrical team from 24% to 100% utilization, and integrated servo-actuated mid-flight payload drops with the Payload team.
- Presented mapping, computer vision, and compute-utilization results to Lockheed Martin engineers at the Preliminary Design Review, earning approval to proceed to the actual mission.

### Embedded Systems Member: Autonomous Ground Vehicle (University Club)

Pomona, CA

Northrop Grumman Collaboration Project

February 2026 – June 2026

- Identified counterfeit silicon by probing a DS2431-class authenticator with a Bus Pirate v4 and logic analyzer, confirming a family-code mismatch and keeping a 20-unit batch of clones out of the build.
- Wrote bare-metal C firmware with direct register control (GPIO, timers, interrupts) plus sensor and actuator drivers for flight hardware, sustaining 1 kHz sensor sampling with sub-10  $\mu$ s interrupt latency.
- Designed and presented the drone-to-rover telemetry workflow to Northrop Grumman engineers at the Preliminary Design Review, earning approval to proceed to CDR.
- Resolved Linux device-communication and connectivity failures using journalctl, dmesg, htop, networking tools, and SSH, cutting field-test downtime from hours to under 30 minutes.

## Skills

**Languages and Tools:** C, C++, Python, Linux, Git, VHDL, ARM Assembly, Bash, bare-metal, UART, SPI, CAN, ROS2, Gazebo, YOLO, PyTorch, computer vision, machine learning, GitHub Actions, Docker, TypeScript, Node.

**Coursework:** Microcontrollers & Embedded Systems, Circuit Analysis, Signals & Systems, Data Structures & Algorithms.

**Languages:** English, Russian, Armenian